



PostProc User's Guide (v.7.5)

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1 Introduction

This guide covers the usage of **PostProc**, version 7.5: an open-source package for postprocessing of data produced by **PWscf** and **CP**.

This guide assumes that you know the physics that **PostProc** describes and the methods it implements. It also assumes that you have already installed, or know how to install, **QUANTUM ESPRESSO**. If not, please read the general User's Guide for **QUANTUM ESPRESSO**, found in subdirectory **Doc/** of the main **QUANTUM ESPRESSO** directory, or consult the web site: <http://www.quantum-espresso.org>.

Further documentation, beyond what is provided in this guide, can be found in the directory **PP/Doc/**, containing a copy of this guide. People who want to contribute to **QUANTUM ESPRESSO** should read the Wiki pages on GitLab: <https://gitlab.com/QEF/q-e/-/wikis>.

2 People and terms of use

The `PostProc` package was originally developed by Stefano Baroni, Stefano de Gironcoli, Andrea Dal Corso (SISSA), Paolo Giannozzi (Univ. Udine), and many others. We mention in particular:

- Dong Yang and Qin Liu (JSG) for calculation of DORI (10.1021/ct500490b) and for spin-polarized ELF;
- Minsu Ghim (Seoul National U.) for Ji Hoon Ryoo's spin-current matrix elements (Phys. Rev. B 99, 235113) for spin Hall conductivity using Wannier interpolation, in `pw2wannier.f90`;
- Yang Jiao, Elsebeth Schröder, Per Hyldgaard (Chalmers) for the `ppacf.x` code;
- Alberto Otero-de-la-Roza for the `pw2critic.x` utility;
- Junfeng Qiao for improvements to `plotband.x`;
- Olivia Pulci, Adriano Mosca Conte, Davide Grassano (RomaII) for the `pw2gw` utility;
- Cyrille Barreteau and Alexander Smogunov (CEA) for magnetic anisotropy with the Force Theorem in `projwfc.f90`;
- Andrea Benassi (SISSA) for the `epsilon` utility, Tae Yun Kim and Cheol-Hwan Park (Seoul National University) for fixes to it;
- Dmitry Korotin (Inst. Met. Phys. Ekaterinburg) for the `wannier_ham` utility;
- Georgy Samsonidze (Bosch Research) for the interface with the Berkeley GW code, Fangzhou Zhao (Berkeley) for its extension to hybrid and meta-GGA functionals;
- The late Prof. Eyvaz Isaev for the Fermi Surface code;
- Natalie Holzwarth (WFU) for the PAW projection in code `projwfc.f90`;
- Takashi Koretsune and Florian Thoele (ETHZ) for noncollinear magnetisation support with USPP and PAW pseudopotentials in code `pw2wannier.f90`.
- Leopold Talirz (U.York) for extensions and fixes to `pp.x`.
- Iurii Timrov (PSI) for the implementation of `wannier2pw.f90`.

`PostProc` is free software, released under the GNU General Public License. See: <http://www.gnu.org/licenses/old-licenses/gpl-2.0.txt>, or the file `License` in the distribution).

We shall greatly appreciate if scientific work done using the QUANTUM ESPRESSO distribution will contain an acknowledgment to the following references:

P. Giannozzi, S. Baroni, N. Bonini, M. Calandra, R. Car, C. Cavazzoni, D. Ceresoli, G. L. Chiarotti, M. Cococcioni, I. Dabo, A. Dal Corso, S. Fabris, G. Fratesi, S. de Gironcoli, R. Gebauer, U. Gerstmann, C. Gougoussis, A. Kokalj, M. Lazzeri, L. Martin-Samos, N. Marzari, F. Mauri, R. Mazzarello, S. Paolini, A. Pasquarello, L. Paulatto, C. Sbraccia, S. Scandolo, G. Sclauzero, A. P. Seitsonen, A. Smogunov, P. Umari, R. M. Wentzcovitch, J.Phys.: Condens.Matter 21, 395502 (2009)

and

P. Giannozzi, O. Andreussi, T. Brumme, O. Bunau, M. Buongiorno Nardelli, M. Calandra, R. Car, C. Cavazzoni, D. Ceresoli, M. Cococcioni, N. Colonna, I. Carnimeo, A. Dal Corso, S. de Gironcoli, P. Delugas, R. A. DiStasio Jr, A. Ferretti, A. Floris, G. Fratesi, G. Fugallo, R. Gebauer, U. Gerstmann, F. Giustino, T. Gorni, J. Jia, M. Kawamura, H.-Y. Ko, A. Kokalj, E. Küçükbenli, M. Lazzeri, M. Marsili, N. Marzari, F. Mauri, N. L. Nguyen, H.-V. Nguyen, A. Otero-de-la-Roza, L. Paulatto, S. Poncé, D. Rocca, R. Sabatini, B. Santra, M. Schlipf, A. P. Seitsonen, A. Smogunov, I. Timrov, T. Thonhauser, P. Umari, N. Vast, X. Wu, S. Baroni, *J.Phys.: Condens.Matter* 29, 465901 (2017)

Users of the GPU-enabled version should also cite the following paper:

P. Giannozzi, O. Baseggio, P. Bonfà, D. Brunato, R. Car, I. Carnimeo, C. Cavazzoni, S. de Gironcoli, P. Delugas, F. Ferrari Ruffino, A. Ferretti, N. Marzari, I. Timrov, A. Urru, S. Baroni, *J. Chem. Phys.* 152, 154105 (2020)

Note the form `QUANTUM ESPRESSO` for textual citations of the code. Please also see package-specific documentation for further recommended citations. Pseudopotentials should be cited as (for instance)

[] We used the pseudopotentials `C.pbe-rrjkus.UPF` and `O.pbe-vbc.UPF` from <http://www.quantum-espresso.org>.

3 Compilation

`PostProc` is part of the `QUANTUM ESPRESSO` distribution and depends upon `PWscf` for compilation. For instruction on how to download and compile `QUANTUM ESPRESSO`, please refer to the general Users' Guide, available in file `Doc/user_guide.pdf` under the main `QUANTUM ESPRESSO` directory, or in web site <http://www.quantum-espresso.org>.

Once `QUANTUM ESPRESSO` is correctly configured, `PostProc` can be compiled by just typing `make pp`, from the main `QUANTUM ESPRESSO` directory; or typing `make` from the `PP/` subdirectory. Several executable codes are produced in `PP/bin` and linked to `bin/`.

4 Usage

All codes for which input documentation is not explicitly mentioned below have some documentation in the header of the fortran sources. In the following, subdirectories containing examples are found in `PP/examples/`; "Example N" stands for subdirectory `PP/examples/exampleN/`.

All quantities whose dimensions are not explicitly specified are in RYDBERG ATOMIC UNITS. Charge is "number" charge (i.e. not multiplied by e); potentials are in energy units (i.e. they are multiplied by e).

4.1 Plotting selected quantities

The main postprocessing code `pp.x` extracts the specified data from the data files produced by `PWscf` (`pw.x` executable) or `CP` (`cp.x` executable); prepares data for plotting by writing them into formats that can be read by several plotting programs.

Quantities that can be read or calculated are:

- charge density
- spin polarization
- various potentials
- local density of states at E_F
- local density of electronic entropy
- STM images
- selected squared wavefunction
- ELF (electron localization function)
- RDG (reduced density gradient)
- integrated local density of states

Various types of plotting (along a line, on a plane, three-dimensional, polar) and output formats (including the popular cube format) can be specified. Moreover data can be saved to an intermediate (formatted) file so that more data sets can be summed or subtracted in a later run. The output files can be directly read by the free plotting system Gnuplot (1D or 2D plots), or by code `plotrho.x` that comes with `PostProc` and produces PostScript 2D plots, or by advanced plotting software XCrySDen (3D plots).

See file `PP/Doc/INPUT_PP.*` for a detailed description of the input for code `pp.x`. See Example 01 for an example of a charge density plot, Example 03 for an example of STM image simulation.

Planar averages Code `plan_avg.x` calculates planar averages of Kohn-Sham orbitals. Input documentation is in the header of `PP/src/plan_avg.f90`.

Code `average.x` calculates planar averages of quantities produced by `pp.x` (e.g. potentials, charge, magnetization densities). Note that `average.x` reads the intermediate file produced by `pp.x`, not data files produced by `pw.x`. Examples of usage of `average.x` can be found in `PP/examples/WorkFct_example/` and in `PP/examples/dipole_example/`.

All-electron charge `pawplot.x` produces plots of the all-electron charge for PAW calculations. Input documentation in the header of `PP/src/pawplot.f90`.

4.2 About Bader's analysis

In <http://theory.cm.utexas.edu/henkelman/code/bader/> one can find a software that performs Bader's analysis starting from charge on a regular grid. One should use PAW to compute the charge density. The required "cube" format can be produced using `pp.x` (info by G. Lapenna who has successfully used this technique, but adds: "Problems occur with polar X-H bonds or in all cases where the zero-flux of density comes too close to atoms described with pseudo-potentials"). This code should perform decomposition into Voronoi polyhedra as well, in place of obsolete code `voronoy.x` (removed from distribution since v.4.2). Alternatively, you can use *CRITIC2*, available at <https://github.com/aoterodelaroza/critic2>, which can read directly `pw.x` output and "XSF" files. *CRITIC2* functionality include Bader's AIM, ELF, laplacian of density and potentials, non-covalente interaction (NCI) plots and much more.

4.3 Band structure, Fermi surface

The code `bands.x` reads data file(s), extracts eigenvalues, regroups them into bands (the algorithm used to order bands and to resolve crossings may not work in all circumstances, though). The output is written to a file in a simple format that can be directly read and converted to plottable format by auxiliary code `plotband.x`. Unpredictable plots may result if k-points are not in sequence along lines, or if two consecutive points are the same. The code `bands.x` performs as well a symmetry analysis of the band structure. For a complete input description, see `Doc/INPUT_bands.*`. See Example 01, Example 04 and Example 06 for simple band plots.

The plotting of Fermi surfaces can be performed using code `fs.x`. The resulting file in `.bxsf` format can be read and plotted using XCrySDen. See Example 02 for an example of Fermi surface visualization (Ni, including the spin-polarized case).

4.4 Projection over atomic states, DOS, projected band structure

The code `projwfc.x` calculates projections of wavefunctions over atomic orbitals. The atomic wavefunctions are those contained in the pseudopotential file(s). The Löwdin population analysis (similar to Mulliken analysis) is presently implemented. The projected DOS (or PDOS: the DOS projected onto atomic orbitals) can also be calculated and written to file(s). More details on the input data are found in file `PP/Doc/INPUT_PROJWFC.*`. The ordering of the various angular momentum components (defined in routine `y1mr2.f90`) is as follows: $P_{0,0}(t)$, $P_{1,0}(t)$, $P_{1,1}(t)\cos\phi$, $P_{1,1}(t)\sin\phi$, $P_{2,0}(t)$, $P_{2,1}(t)\cos\phi$, $P_{2,1}(t)\sin\phi$, $P_{2,2}(t)\cos2\phi$, $P_{2,2}(t)\sin2\phi$ and so on, where $P_{l,m}$ =Legendre Polynomials, $t = \cos\theta = z/r$, $\phi = \text{atan}(y/x)$.

Data produced by code `projwfc.x` can be further analysed using auxiliary codes `sumpdos.x` (sums selected PDOS by specifying the names of files containing the desired PDOS: type `sumpdos.x -h` or look into the source code for more details) and `plotproj.x`. A more sophisticated tool is the script `PP/tools/sum_states.py`, by Julien Larrucea: documentation in http://larrucea.eu/sum_states-py-2/.

The total electronic DOS can also be calculated by code `dos.x`, whose complete input documentation is in `PP/Doc/INPUT_DOS.*` See Example 02 for total and projected electronic DOS calculations, -and for projected band structure; see Example 03 for projected and local DOS calculations.

The DOS projected over *molecular* states (e.g. for a molecule on a surface system) can be computed using code `molecularpdos.x` (courtesy of Guido Fratesi). See file `PP/Doc/INPUT_MOLDOS.*` for input documentation and directory `PP/examples/MolDos_example/` for an example.

The calculation of magnetic anisotropy using the Force Theorem is described in the following paper: <https://journals.aps.org/prb/abstract/10.1103/PhysRevB.90.205409>. An example and a README can be found in `PP/examples/ForceTheorem_example/`

4.5 Color plot of the Fermi velocity and the orbital character on Fermi surfaces

You can plot any quantity on Fermi surfaces as a color plot by using `fermisurfer` program¹. `fermi_velocity.x` and `fermi_proj.x` are used to generate an input file for `fermisurfer` from the output of `pw.x` or `projwfc.x`.

`fermi_velocity.x` generates a color-plot of Fermi velocity. You use it as follows:

¹<http://osdn.jp/projects/fermisurfer/>

1. Run pw.x with K_POINT automatic.
2. Run

```
$ fermi_velocity.x -in {pw.x input file}
```

3. vfermi.frmsf is generated

fermi_proj.x generates a color plot of an orbital character. You use it as follows:

1. Run pw.x with K_POINT automatic.
2. Run projwfc.x just to generate {prefix}.save/atomic_proj.*.
3. Run

```
$ fermi_proj.x -in {input file}
```

Input-file format is as follows:

```
&PROJWFC
  {The same as the input of projwfc.x}
/
{Number of target wavefunctions}
{Index of target WFC1} {Index of target WFC2} {Index of target WFC3} ...
```

It generates $\sum_{i=1}^{n_{\text{target}}} |\langle \varphi_{\text{target}(i)}^{\text{atom}} | \varphi_{nk} \rangle|^2$, where n_s and $\text{target}(i)$ are the number of the target wavefunctions and the indices of target wavefunctions, respectively.

4. The above quantity is written into "proj.frmsf", which can be read by FermiSurfer program.

There is an example of fermi_velocity.x and fermi_proj.x in fermisurf_example/.

4.6 Wannier functions

There are several Wannier-related utilities in PostProc:

1. The "Poor Man Wannier" code pmw.x, to be used in conjunction with DFT+U calculations: see Example 05.
2. The interface with Wannier90 code, pw2wannier.x: see the documentation in W90/ (you may install the Wannier90 plug-in via `make w90`). For spin-current matrix elements, implemented in routine `compute_shc`: "it writes .slu and .shu files used for WANNIER-BERRI (<https://github.com/stepan-tsirkin/wannier-berri/>), and also will be utilized through postw90.x (<https://github.com/manxkim/wannier90/tree/SHC/src>) in Wannier90. In WANNIER-BERRI, .shu and .slu files can be used to calculate the quantity "opt.SHCryo". In Wannier90, add "berry_task = shc" and "shc_ryoo=.true.". in the input parameters of postw90.x. They activate the calculation of SHC using .shu and .slu."

3. The `wannier_ham.x` code generates a model Hamiltonian in Wannier functions basis: see `PP/examples/WannierHam_example/`.
4. The interface with Wannier90 code, `wannier2pw.x`: it builds Wannier functions as Hubbard projectors for DFT+U

Note that the `wfdd.x` code has been moved to CP.

4.7 Interfaces to/from other code

Codes `pw2bgw.x` convert data files from `pw.x` to a format suitable for usage by the Berkeley GW code. See file `Doc/INPUT_pw2bgw.*` for input data documentation. Code `bgw2pw.x`, performing the inverse conversion, no longer works: a copy that worked for the old file format is kept for reference in `bgw2pw.f90.orig`.

Code `pw2gw.x` converts data files from `pw.x` to a format suitable for usage by another GW code, computes optical properties in single-particle approach (Fermi Golden Rule). See file `Doc/INPUT_pw2gw.html` for input data documentation, directory `pw2gw_example/` for an example of usage.

Code `open_grid.x` writes Kohn-Sham orbitals for the complete k-point grid (not symmetry-independent points only) in real space. Useful for further processing. It can be used to generate the Kohn-Sham state data required in `pw2wannier.x` and Wannier90 from the initial SCF calculation, bypassing the non-SCF calculation step.

Code `pw2critic.x` is an interface to the CRITIC2 code by Alberto Otero-de-la-Roza. This program creates a `pwc` file containing the Kohn-Sham orbitals from an SCF calculation (or from the output of `open_grid.x`). These orbitals are used for post-processing in CRITIC2.

Code `pw_export.f90` no longer works and is no longer present.

4.8 Other tools

Exchange-correlation Code `ppacf.x` computes the coupling constant dependency of the exchange correlation potential $E_{xc,\lambda}$, $\lambda \in [0 : 1]$ and the spatial distribution of the exchange-correlation energy density and kinetic correlation energy density according to: Y. Jiao, E. Schröder, and P. Hyldgaard, Phys. Rev. B 97, 085115 (2018). See `PP/Doc/INPUT_PPACF.html`.

Wavefunction conversion Code `wfck2r.x` converts Kohn-Sham orbitals from reciprocal to real space. It is a useful starting point if you need to access wavefunctions and perform postprocessing operations that are not implemented in QUANTUM ESPRESSO.

Dielectric function Code `epsilon.x` calculates RPA frequency-dependent complex dielectric function. Documentation is in file `Doc/eps_man.tex`.

Core-level shifts Code `initial_state.x` calculates the initial state contribution to the Core-level shift. See `CLS_IS_example/` for an example, and `CLS_FS_example/` for the corresponding final state calculation of Core-level shifts.

5 Troubleshooting

Almost all problems in QUANTUM ESPRESSO arise from incorrect input data and result in error stops. Error messages should be self-explanatory, but unfortunately this is not always true. If the code issues a warning messages and continues, pay attention to it but do not assume that something is necessarily wrong in your calculation: most warning messages signal harmless problems.

Some postprocessing codes complain that they do not find some files Most likely you are not reading the correct data files, or you are not following the correct procedure for postprocessing.

For Linux PC clusters in parallel execution: in at least some versions of MPICH, the current directory is set to the directory where the executable code resides, instead of being set to the directory where the code is executed. This MPICH weirdness may cause unexpected failures in some postprocessing codes that expect a data file in the current directory. Workaround: use symbolic links, or copy the executable to the current directory.